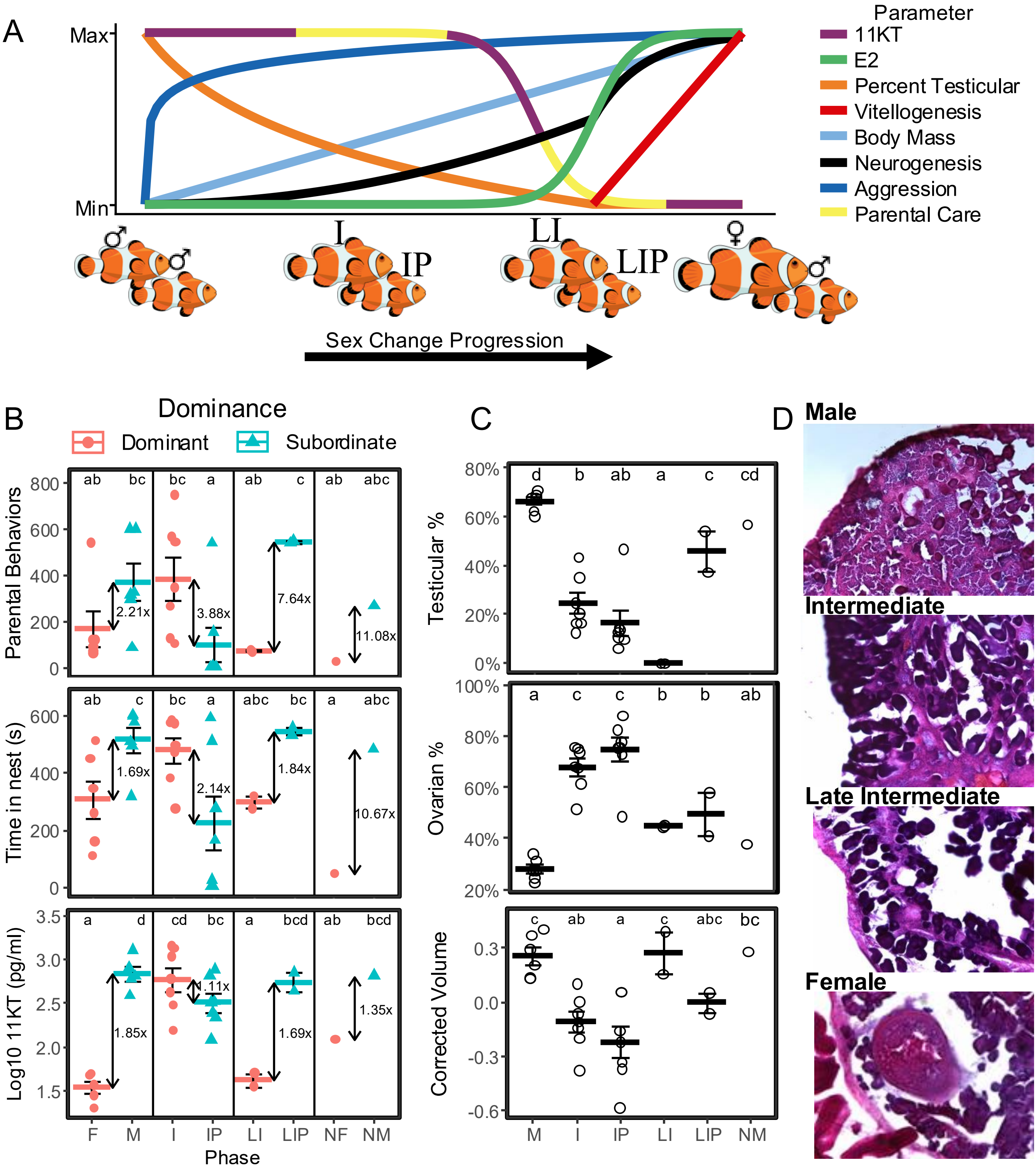


A
Max
Min
Parameter
11KT
E2
Percent Testicular
Vitellogenesis
Body Mass
Neurogenesis
Aggression
Parental Care
♂
I
IP
LI
LIP
♀
Sex Change Progression
B
Dominance
Dominant
Subordinate
Parental Behaviors
Time in nest (s)
Log10 11KT (pg/ml)
2.21x
3.88x
7.64x
11.08x
1.69x
2.14x
1.84x
10.67x
1.85x
1.11x
1.69x
1.35x
F M I IP LI LIP NF NM
Phase
C
Testicular %
Ovarian %
Corrected Volume
M I IP LI LIP NM
D
Male
Intermediate
Late Intermediate
Female

Fig.1. Male, intermediate, and female anemonefish display distinct physiology and behavior. (A) Schematic outlining the behavioral and physiological changes that occur during protandrous sex change in anemonefish. (B) Top panel: Parental behaviors observed during a parental care assay. Middle panel: Time spent in the nest during parental care assay; a proxy for parental care. Bottom Panel: Log10 transformed serum 11-ketotestosterone. (C)Top panel: Percent of the gonad made of testicular tissue. Middle panel: Percent of the gonad made of ovarian tissue. Bottom panel: Gonadal volume corrected for the individual's length (methods). (D) Example hematoxylin and eosin stains of gonads from the major phases of sex change. In B, plots are subdivided such that the dominant and subordinate individuals from each pair appear together reflected by shape and color.. Arrows indicate fold-difference between the two groups. In B, C, crossbar indicates mean, error bars indicate standard error of the mean (SEM). Letters indicate significance: groups that do not share a letter are significantly different.

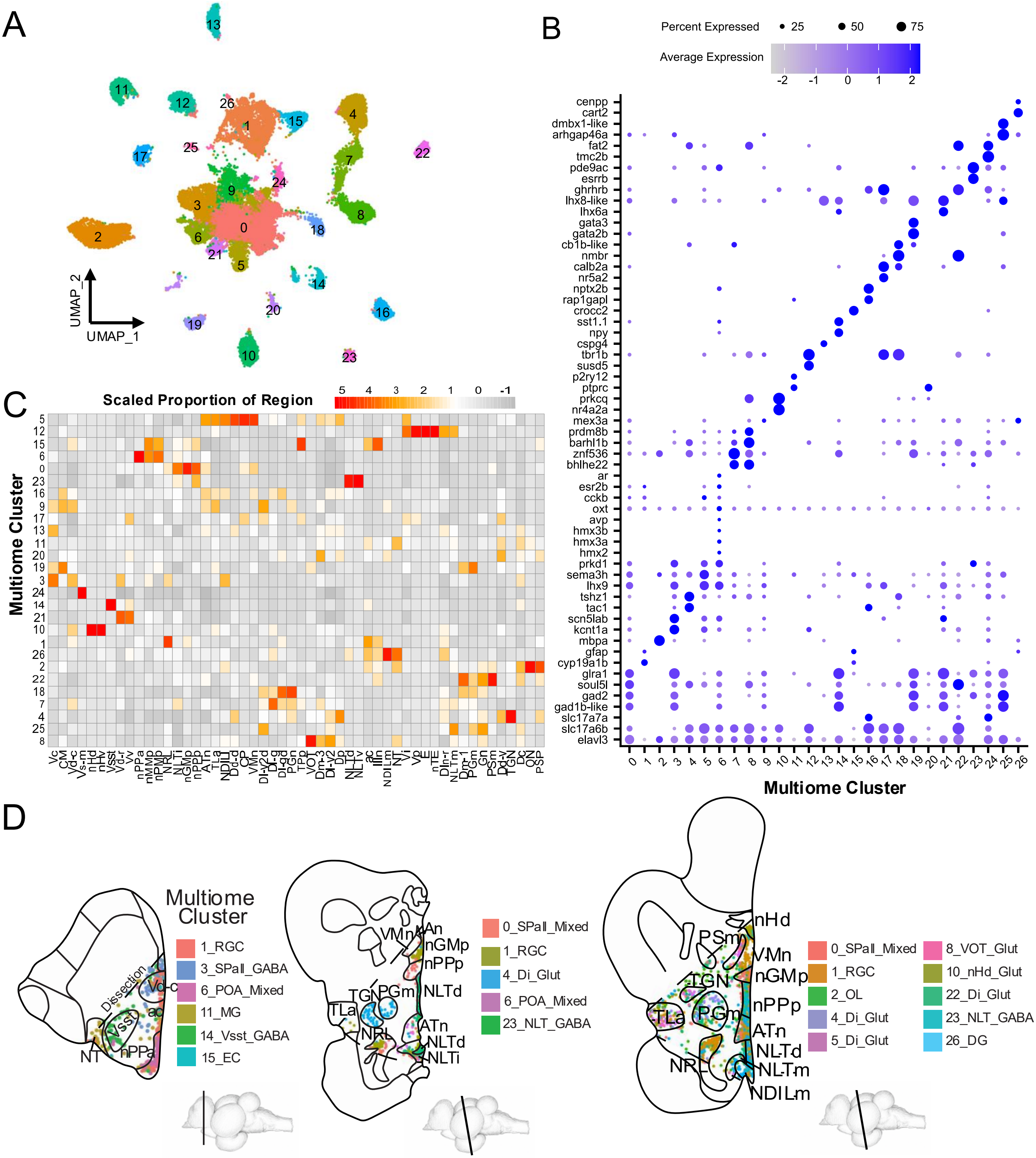
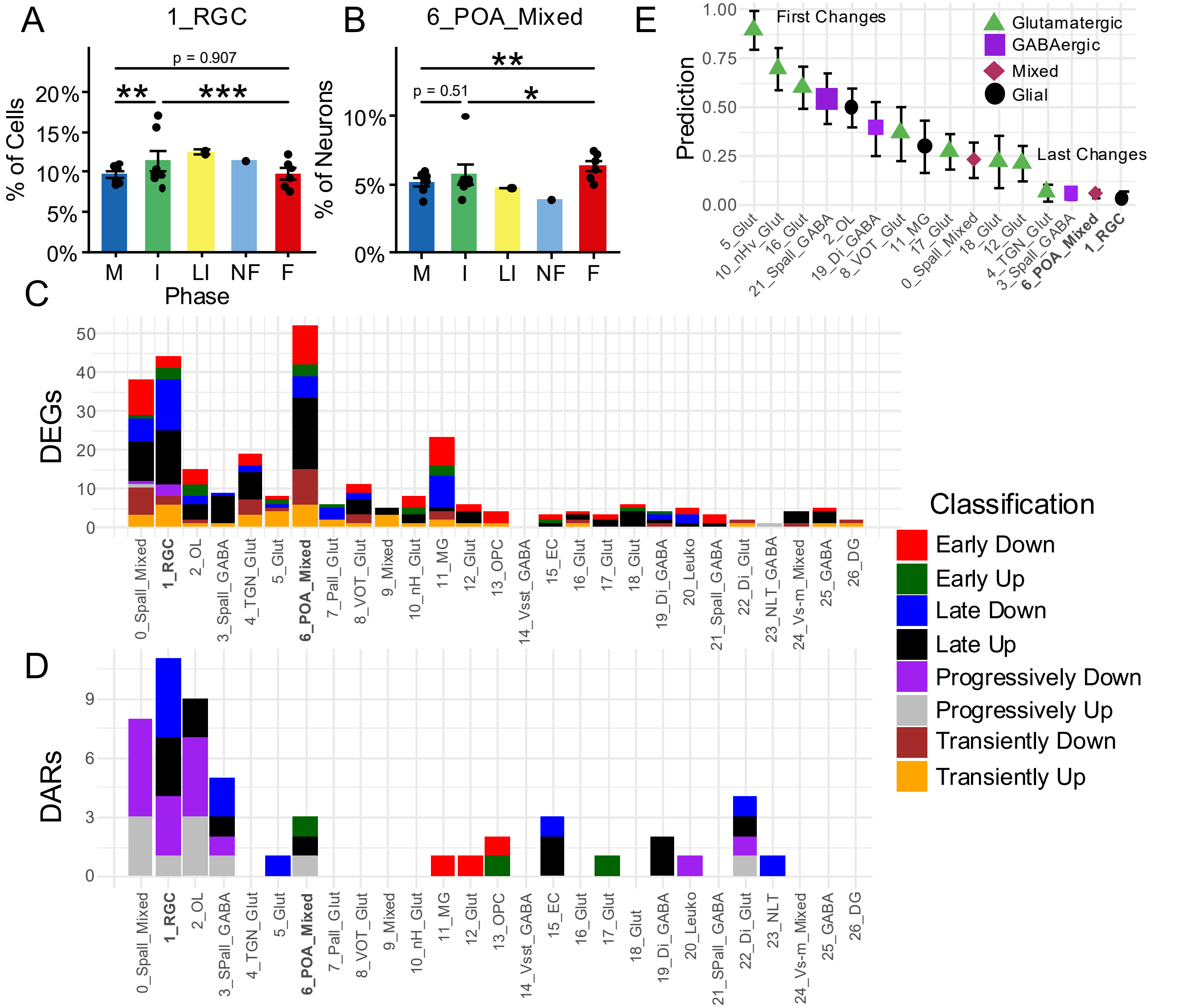

A
UMAP_1
UMAP_2
B
Percent Expressed
25
50
75
Average Expression
-2
-1
0
1
2
cenpp
cart2
dmbx1-like
arhgap46a
fat2
tmc2b
pde9ac
esrrb
ghrhrb
lhx8-like
lhx6a
gata3
gata2b
cb1b-like
nmbr
calb2a
nr5a2
nptx2b
rap1gapl
crocc2
sst1.1
npy
cspg4
tbr1b
susd5
p2ry12
ptprc
prkcq
nr4a2a
mex3a
prdm8b
barhl1b
znf536
bhlhe22
ar
esr2b
cckb
oxt
avp
hmx3b
hmx3a
hmx2
prkd1
sema3h
lhx9
tshz1
tac1
scn5lab
kcnt1a
mbpa
gfap
cyp19a1b
glra1
soul5l
gad2
gad1b-like
slc17a7a
slc17a6b
elavl3
Multiome Cluster
C
Scaled Proportion of Region
5
4
3
2
1
0
-1
Multiome Cluster
Vc
CM
Vd-c
Vs-m
nHd
nHv
Vsst
Vd-r
Vv
nPPa
nMMp
nPMp
NRL
NLTi
nGMp
nPPp
ATn
TLa
NDILl
Dd-d
CP
VMn
Dl-v2d
Dl-g
Dl-gd
PGn
TPp
VOT
Dm-3
Dl-v2
Dp
NLTd
NLTv
ac
lllfn
NDILm
NT
Vi
Vp
E
nTE
Dln-r
NLTm
Dm-1
PGm
Gn
PSm
Dd-v
TGN
Dc
ON
PSP
D
Multiome Cluster
1_RGC
3_SPall_GABA
6_POA_Mixed
11_MG
14_Vsst_GABA
15_EC
Dissection
Vd-c
ac
Vsst
NT
nPPa
VMn
An
nGMp
nPPp
TGN
PGm
NLTd
TLa
NRL
ATn
NLTd
NLTi
0_SPall_Mixed
1_RGC
4_Di_Glut
6_POA_Mixed
23_NLT_GABA
nHd
PSm
VMn
TGN
nGMp
TLa
PGm
nPPp
ATn
NLTd
NRL
NLTm
NDILm
0_SPall_Mixed
1_RGC
2_OL
4_Di_Glut
5_Di_Glut
8_VOT_Glut
10_nHd_Glut
22_Di_Glut
23_NLT_GABA
26_DG

Fig.2. Multiomics capture high quality transcriptomes, epigenomes, and spatial transcirptomes, (A) UMAP embeddings of our sn-Multiome data with each cluster numbered. (B) Dotplot showing canonical and data-driven markers for each of our multiome clusters. (C) Heatmap showing the scaled proportion of each neuroanatomical region made of each multiome cluster (see supplement xx for abbreviations). (D) Example projections of multiome clusters onto representative brain sections. The dashed line indicates approximately where we dissected brains for sn-multiomics analysis. Below each section is a sketch of an *A. ocellaris* brain with a line indicating the approximate sectioning plane for the given section.



A
1_RGC
p = 0.907
**
***
% of Cells
20%
15%
10%
5%
0%
M
I
LI
NF
F
Phase
B
6_POA_Mixed
**
p = 0.51
*
% of Neurons
10%
5%
0%
E
First Changes
Last Changes
Prediction
1.00
0.75
0.50
0.25
0.00
Glutamatergic
GABAergic
Mixed
Glial
5_Glut
10_nHv_Glut
16_Glut
21_Spall_GABA
2_OL
19_Di_GABA
8_VOT_Glut
11_MG
17_Glut
0_Spall_Mixed
18_Glut
12_Glut
4_TGN_Glut
3_Spall_GABA
6_POA_Mixed
1_RGC
C
DEGs
50
40
30
20
10
0
0_Spall_Mixed
1_RGC
2_OL
3_Spall_GABA
4_TGN_Glut
5_Glut
6_POA_Mixed
7_Pall_Glut
8_VOT_Glut
9_Mixed
10_nH_Glut
11_MG
12_Glut
13_OPC
14_Vsst_GABA
15_EC
16_Glut
17_Glut
18_Glut
19_Di_GABA
20_Leuko
21_Spall_GABA
22_Di_Glut
23_NLT_GABA
24_Vs-m_Mixed
25_GABA
26_DG
D
DARs
9
6
3
0
0_Spall_Mixed
1_RGC
2_OL
3_SPall_GABA
4_TGN_Glut
5_Glut
6_POA_Mixed
7_Pall_Glut
8_VOT_Glut
9_Mixed
10_nH_Glut
11_MG
12_Glut
13_OPC
14_Vsst_GABA
15_EC
16_Glut
17_Glut
18_Glut
19_Di_GABA
20_Leuko
21_SPall_GABA
22_Di_Glut
23_NLT
24_Vs-m_Mixed
25_GABA
26_DG
Classification
Early Down
Early Up
Late Down
Late Up
Progressively Down
Progressively Up
Transiently Down
Transiently Up

Fig.3. Cells change composition, transcriptome, and epigenome asynchronously during sex change. (A) The percent of all cells in our dataset belonging to 1_RGC. (B) The percent of all neurons in our dataset belonging to 6_POA_Mixed. (C) The number of differentially expressed genes (DEGs) by multiome cluster. (D) The number of differentially accessible regions (DARs) by multiome cluster. (E) Machine learning predictions for the mean sex of intermediates based on their expression of DEGs and DARs by multiome cluster; a proxy for the temporal order of sex change Shape and color represents broad cell type. In C, D, bars are colored based on our classification for the manner in which they change sex (see methods). In A, B, points indicate the proportion for each individual. Asterisks: "*": $p < 0.05$, "**": $p < 0.01$, "***": $p < 0.001$.

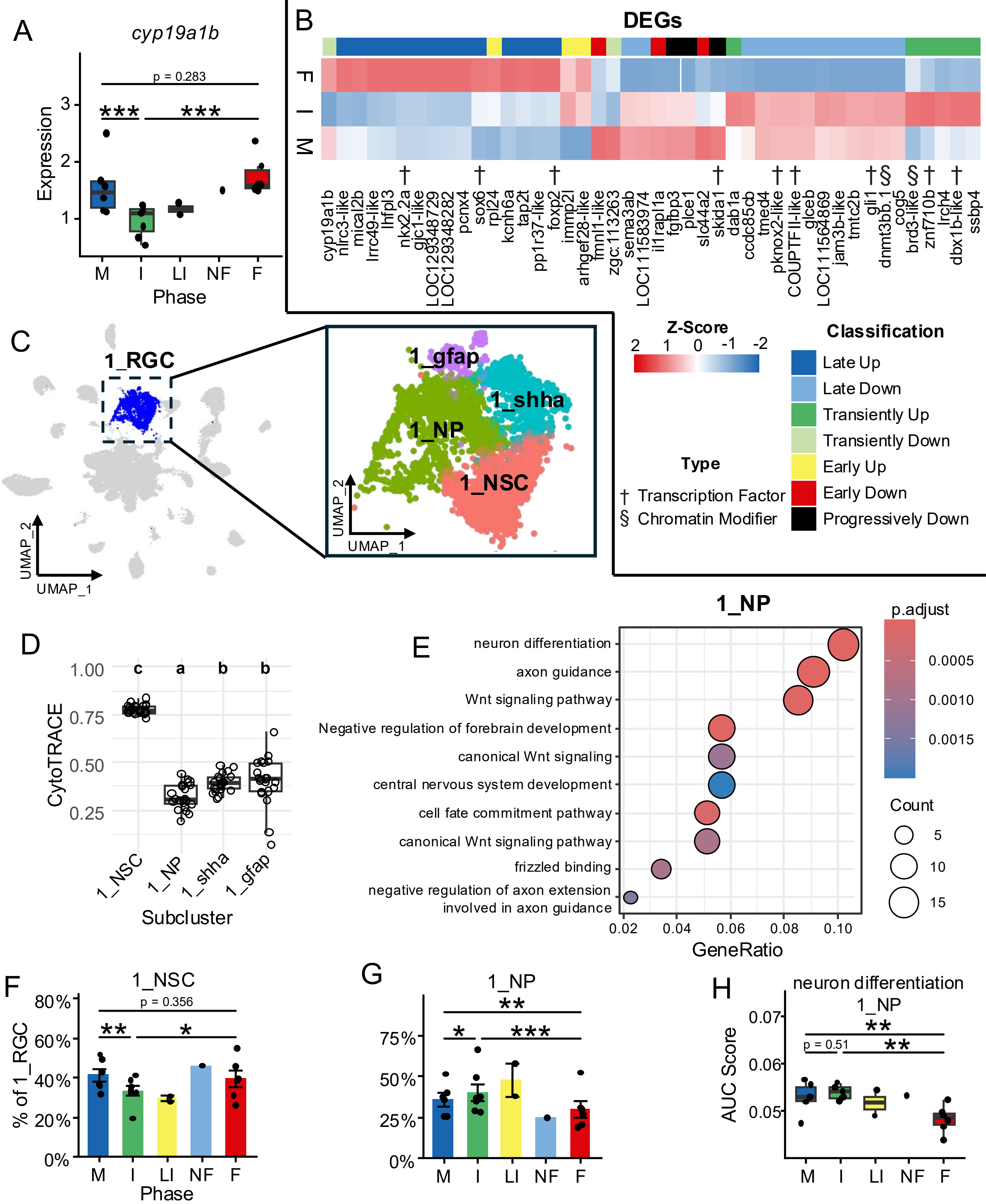
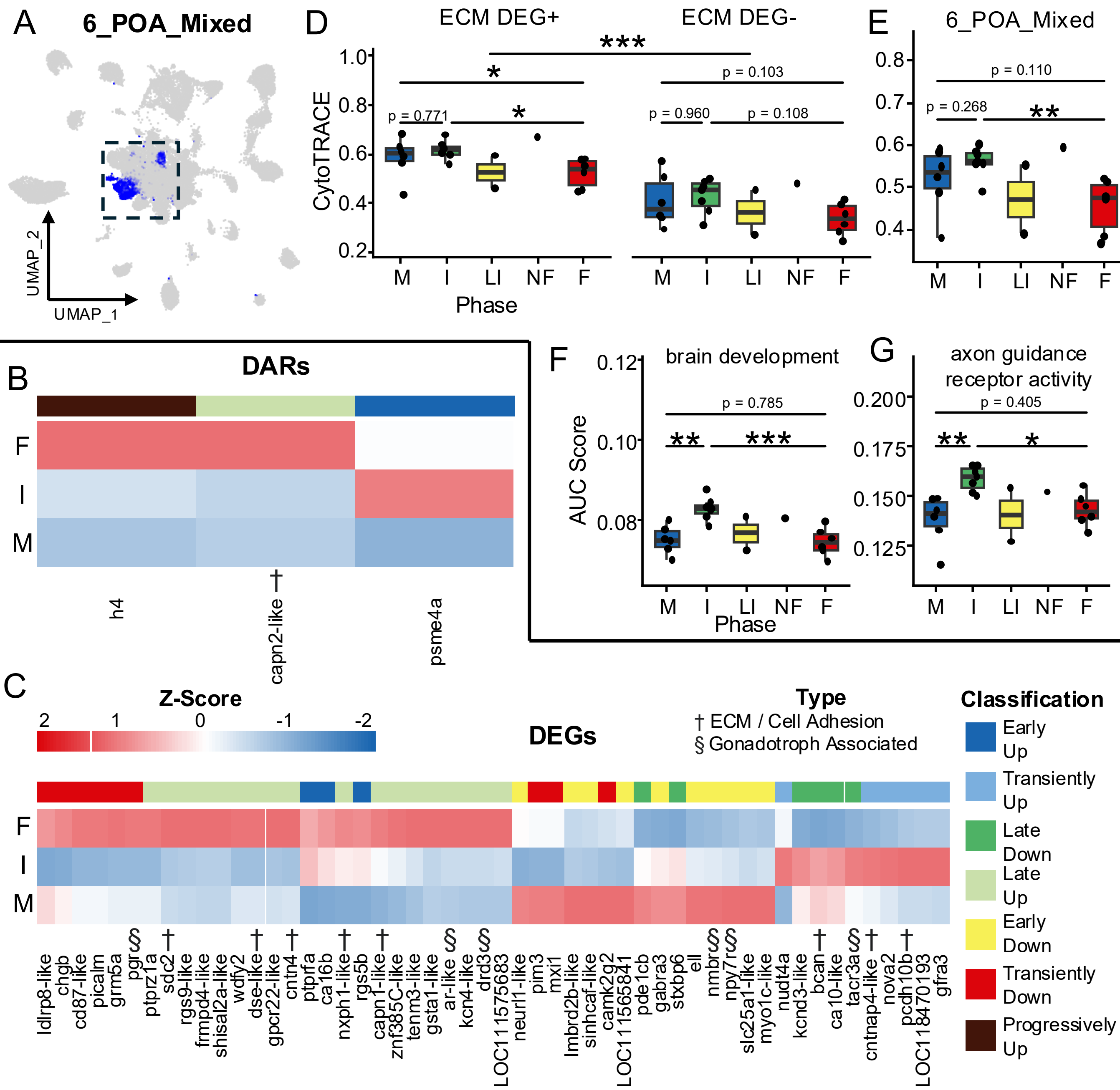
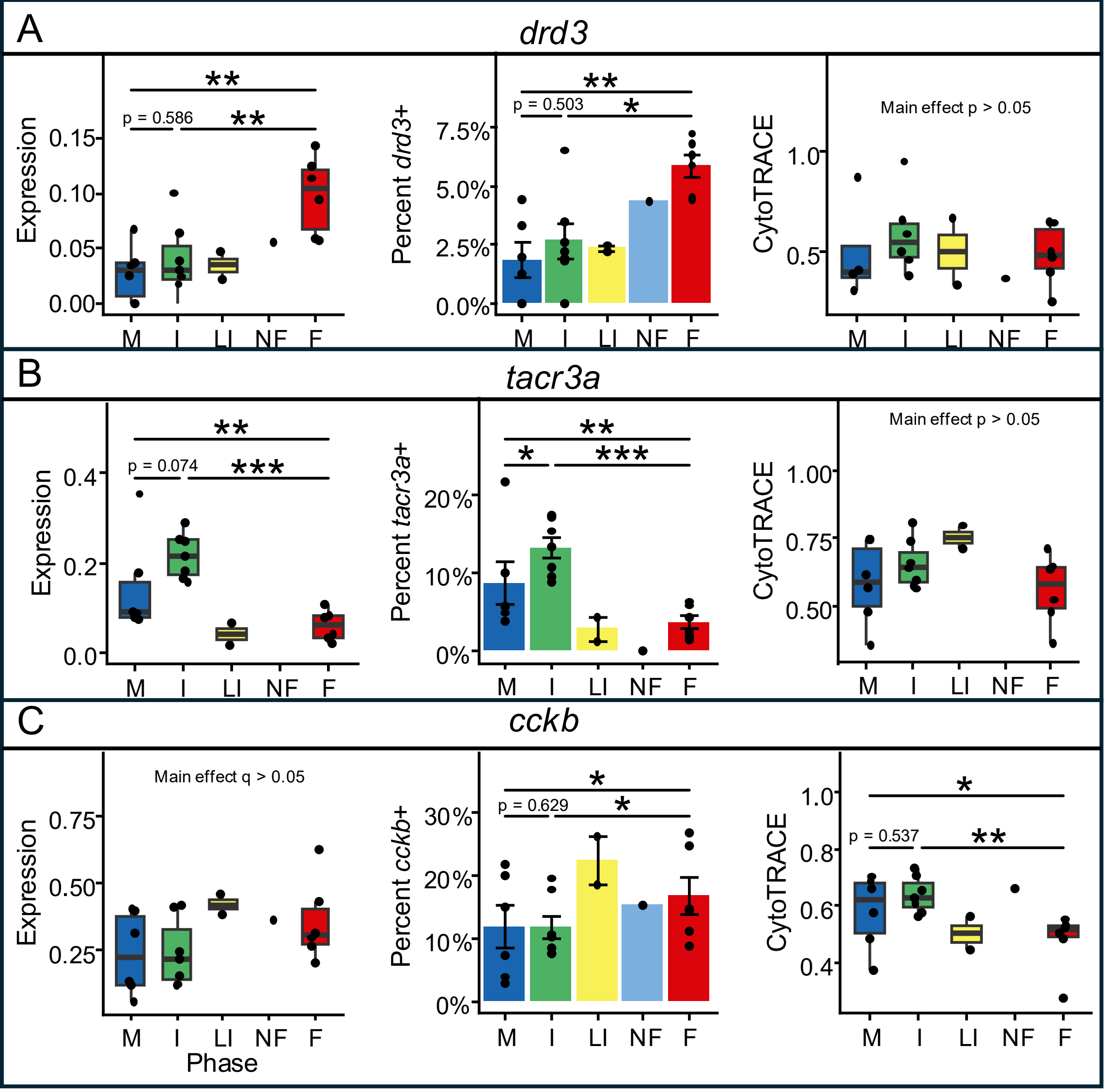

A
cyp19a1b
Expression
Phase
M I LI NF F
p = 0.283
***
B
DEGs
F
I
M
cyp19a1b nlrc3-like mical2b lrrc49-like lhfpl3 nkx2.2a gjc1-like LOC129348729 LOC129348282 pcnx4 sox6 rpl24 kcnh6a tap2t pp1r37-like foxp2 immp2l arhgef28-like fmnl1-like zgc:113263 sema3ab LOC111583974 il1rapl1a fgfbp3 plce1 slc44a2 skida1 dab1a ccdc85cb tmed4 pknox2-like COUPTFII-like glceb LOC111564869 jam3b-like tmtc2b gli1 dnmt3bb.1 cog5 brd3-like znf710b lrch4 dbx1b-like ssbp4
Z-Score
2 1 0 -1 -2
Classification
Late Up
Late Down
Transiently Up
Transiently Down
Early Up
Early Down
Progressively Down
Type
† Transcription Factor
§ Chromatin Modifier
C
1_RGC
UMAP_1
UMAP_2
1_gfap
1_shha
1_NP
1_NSC
D
CytoTRACE
Subcluster
1_NSC 1_NP 1_shha 1_gfap
c a b b
E
1_NP
neuron differentiation
axon guidance
Wnt signaling pathway
Negative regulation of forebrain development
canonical Wnt signaling
central nervous system development
cell fate commitment pathway
canonical Wnt signaling pathway
frizzled binding
negative regulation of axon extension involved in axon guidance
GeneRatio
p.adjust
Count
F
1_NSC
% of 1_RGC
p = 0.356
G
1_NP
H
neuron differentiation
1_NP
AUC Score
p = 0.51

Fig.4. Radial glial cells undergo changes in transcriptome and subtype composition over the course of sex change. (A) Brain aromatase (*cyp19a1b*) expression in 1_RGC. (B) A heatmap showing all DEGs by phase in 1_RGC. Cells are colored by z-score of the mean expression for each phase. The top bar indicates the classification for how each gene changes sex, and the symbols below the matrix indicate what kind of gene each DEG is. (C) UMAP showing subclustering of 1_RGC. (D) Mean CytoTRACE score for each individual by subcluster. A higher score indicates a less mature cell type. Significance is indicated by letters, subclusters who do not share a letter are significantly different from eachother. (E) Gene ontology (GO) analysis for the marker genes of 1_NP. Color indicates false discovery rate adjusted p value, point size indicates the number of genes annotated with that term, and "GeneRatio" indicates the proportion of genes annotated with that GO term divided by all annotated genes in our list. (F) Percent of 1_RGC composed of 1_NSC. (G) Percent of 1_RGC composed of 1_NP. (H) AUCell score of a gene expression module composed of genes labeled with the GO term "neuron differentiation" in 1_NP. (I) Schematic describing our model for how neurogenesis changed over the course of sex change. In A, F-H, points indicate the mean for each individual. Asterisks: "*": $p < 0.05$, "**": $p < 0.01$, "***": $p < 0.001$.



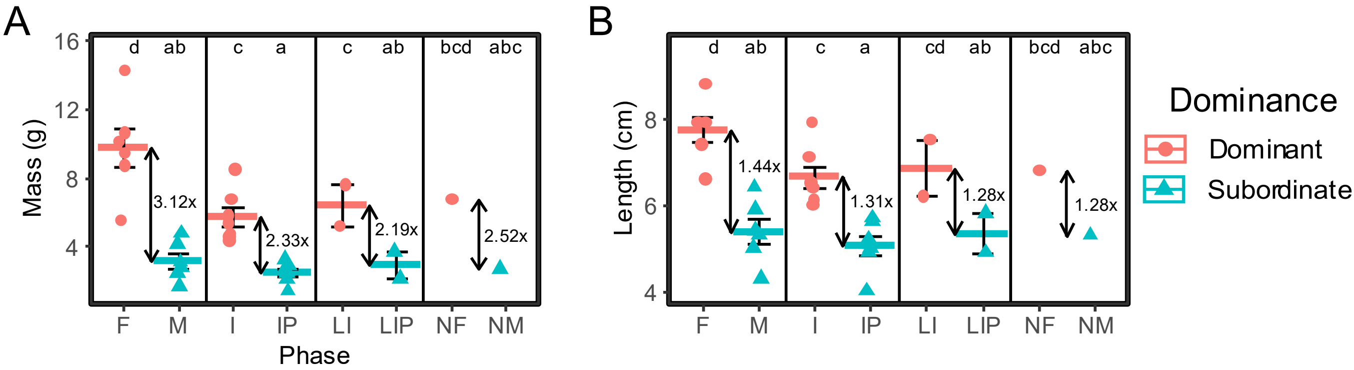
A
6_POA_Mixed
UMAP_2
UMAP_1
D
ECM DEG+
ECM DEG-
CytoTRACE
Phase
p = 0.771
p = 0.960
p = 0.108
p = 0.103
E
6_POA_Mixed
p = 0.268
p = 0.110
B
DARs
h4
capn2-like
psme4a
F
brain development
AUC Score
Phase
p = 0.785
G
axon guidance receptor activity
p = 0.405
C
Z-Score
DEGs
Type
† ECM / Cell Adhesion
§ Gonadotroph Associated
Classification
Early Up
Transiently Up
Late Down
Late Up
Early Down
Transiently Down
Progressively Up
ldlrp8-like
chgb
cd87-like
picalm
grm5a
pgr
ptprz1a
sdc2
rgs9-like
frmpd4-like
shisal2a-like
wdfy2
dse-like
gpcr22-like
cntn4
ptprfa
ca16b
nxph1-like
rgs5b
capn1-like
znf385C-like
tenm3-like
gsta1-like
ar-like
kcn4-like
drd3
LOC111575683
neurl1-like
pim3
mxi1
lmbrd2b-like
sinhcaf-like
camk2g2
LOC111565841
pde1cb
gabra3
stxbp6
ell
nmbr
npy7r
slc25a1-like
myo1c-like
nudt4a
kcnd3-like
bcan
ca10-like
tacr3a
cntnap4-like
nova2
pcdh10b
LOC118470193
gfra3

Fig.5. 6_POA_Mixed undergoes changes in transcriptome and epigenome consistent with neurogenesis. (A) UMAP projection with 6_POA_Mixed highlighted. (B) DARs in 6_POA_mixed. (C) DEGs in 6_POA_Mixed. In B, C, cells are colored by z-score of the mean accessibility or expression for each phase. The top bar indicates the classification for how each gene or region changes sex, and the symbols below the matrix indicate what kind of gene is differentially accessible or differentially expressed. (D) CytoTRACE score by phase split by whether the cell expresses a DEG associated with the extracellular matrix (ECM-DEG) or not. (E) CytoTRACE score for 6_POA_Mixed as a whole. (F) AUCell score for a gene expression module made of genes annotated with the GO term “brain development”. (G) AUCell score for a gene expression module made of genes annotated with the GO term “axon guidance receptor activity”. In D-G, points indicate the mean for each individual. Asterisks: “*”: $p < 0.05$, “**”: $p < 0.01$, “***”: $p < 0.001$.



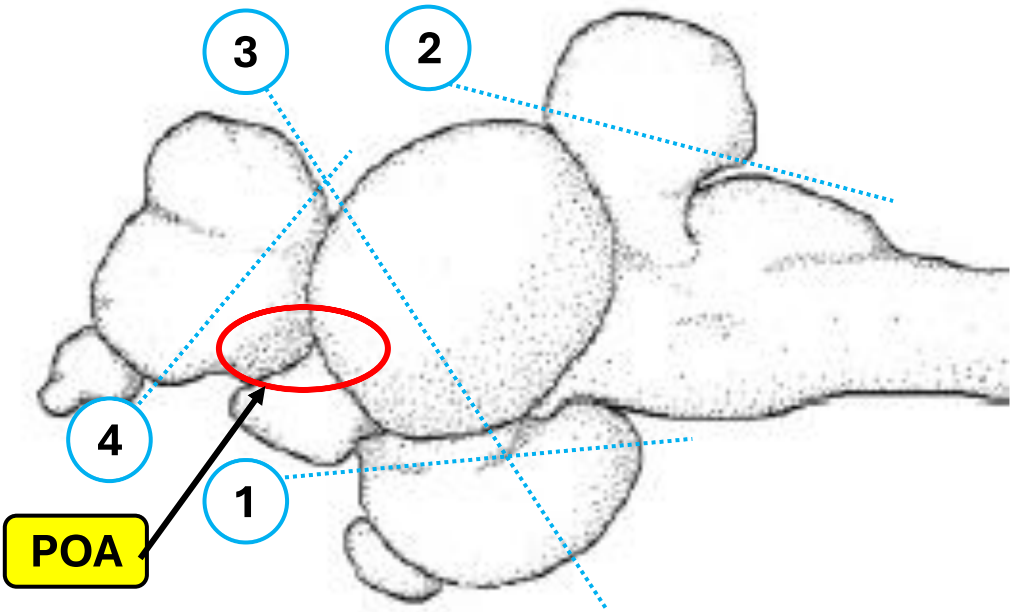
A
drd3
Expression
p = 0.586
**
**
Percent drd3+
p = 0.503
*
**
CytoTRACE
Main effect p > 0.05
M
I
LI
NF
F
B
tacr3a
Expression
p = 0.074
***
**
Percent tacr3a+
*
***
**
CytoTRACE
Main effect p > 0.05
C
cckb
Expression
Main effect q > 0.05
Phase
Percent cckb+
p = 0.629
*
*
CytoTRACE
p = 0.537
**
*

Fig.6. Changes to 6_POA_Mixed Drive Differential Regulation of Gonadotrophs. (A) Left panel expression of *drd3* by phase. Middle panel: Percent of 6_POA_Mixed cells expressing *drd3* by phase. Right panel: CyoTRACE score for *drd3*+ cells by phase.(B) Left panel expression of *tacr3a* by phase. Middle panel: Percent of 6_POA_Mixed cells expressing *tacr3a* by phase. Right panel: CyoTRACE score for *tacr3a*+ cells by phase.(C) Left panel expression of *cckb* by phase. Middle panel: Percent of 6_POA_Mixed cells expressing *cckb* by phase. Right panel: CyoTRACE score for *cckb* + cells by phase. Points indicate the mean for each individual. Asterisks: "*": $p < 0.05$, "**": $p < 0.01$, "***": $p < 0.001$. "Main effect Main effect $p > 0.05$" indicates type III tests were not significant so pairwise comparisons were not computed.



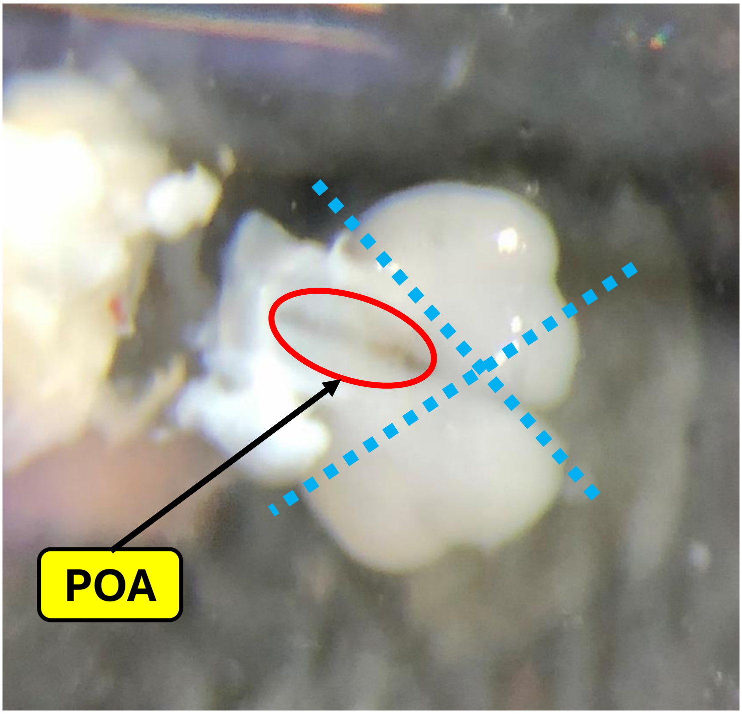
Supplementary Fig.1. (A) Body mass by phase. (B) Body length by phase. Plots are subdivided such that the dominant and subordinate individuals from each pair appear together reflected by shape and color.. Arrows indicate fold-difference between the two groups. Crossbar indicates mean, error bars indicate standard error of the mean (SEM). Letters indicate significance: groups that do not share a letter are significantly different..

A



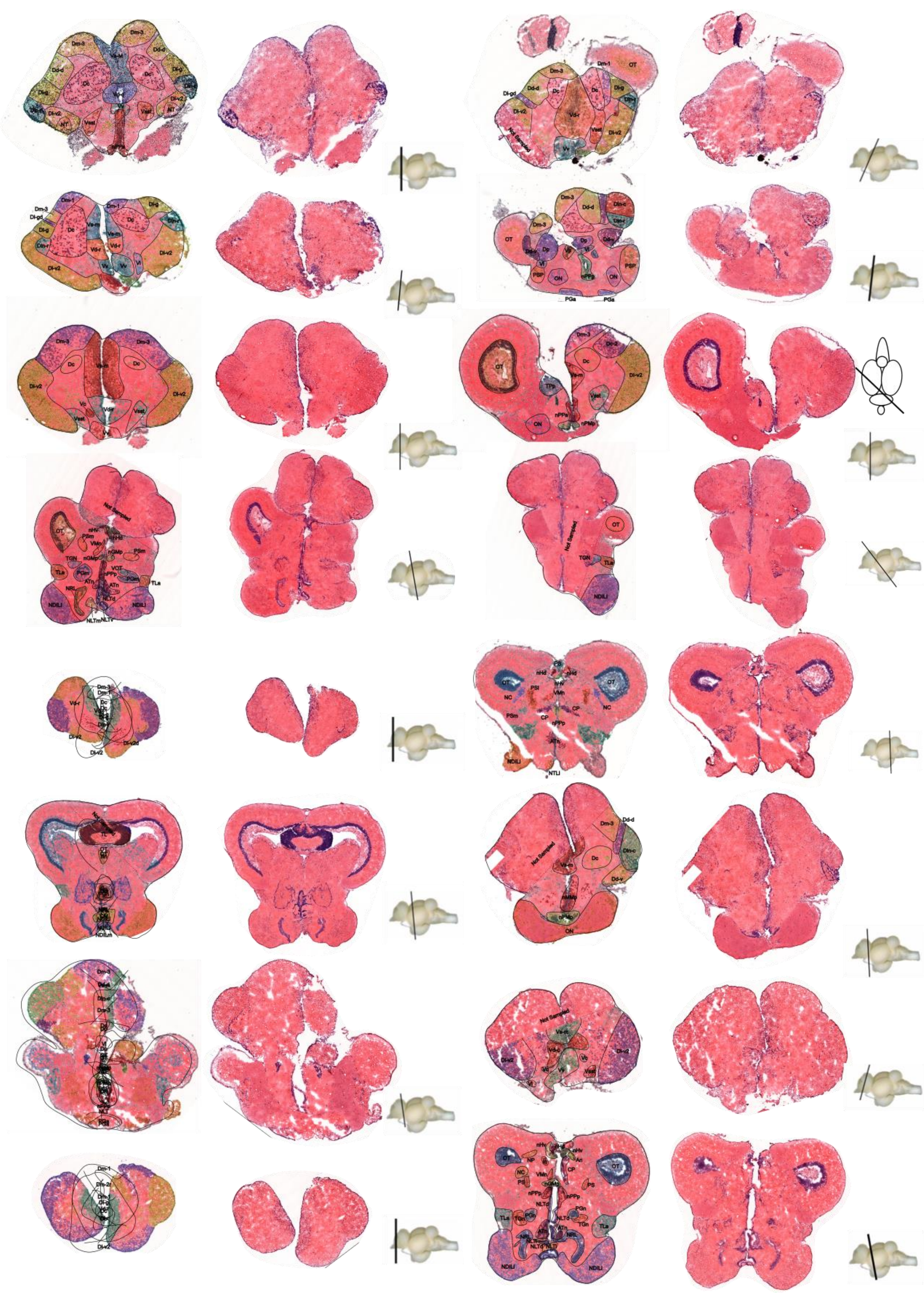
Blue dotted lines represent cuts made

B

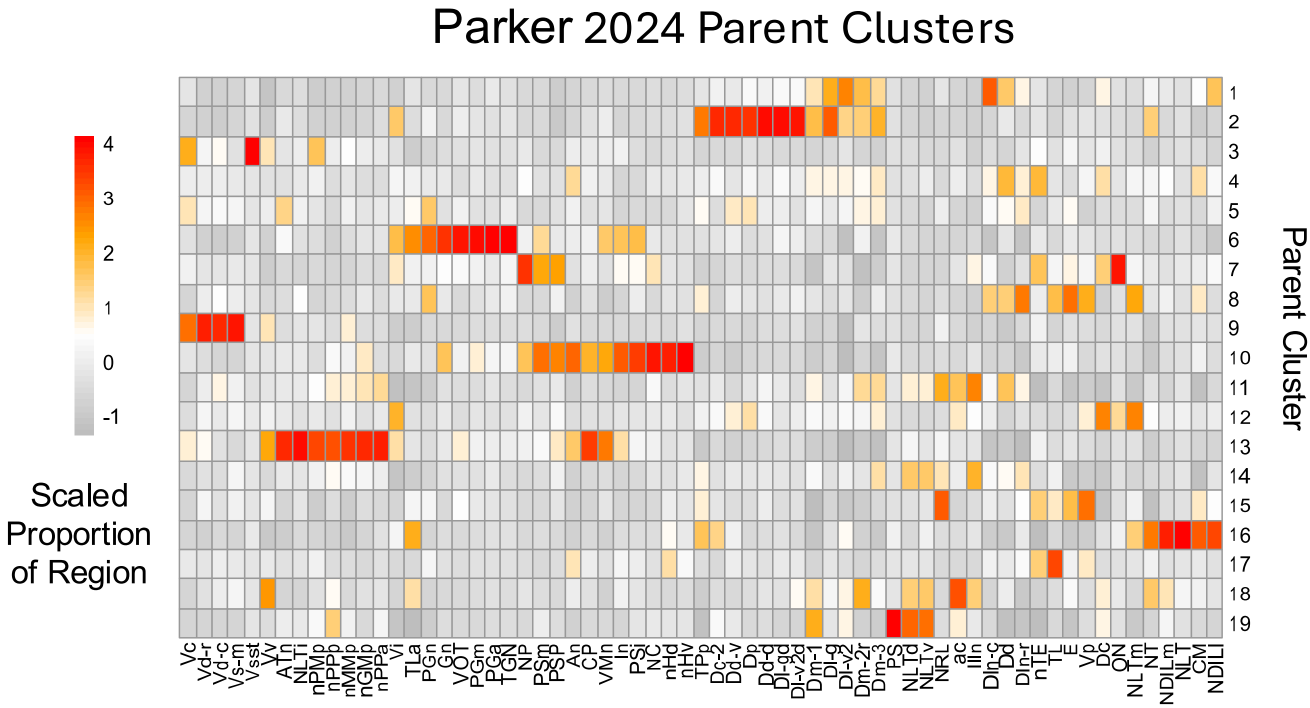


Supplementary Fig.2. (A) Side view of anemonefish brain with dissections indicated by dashes blue lines. The number indicates the order of cuts. (B) Top down vies of anemonefish brain with dissections indicated by dashed blue lines. In A and B, the POA is highlighted with a red circle

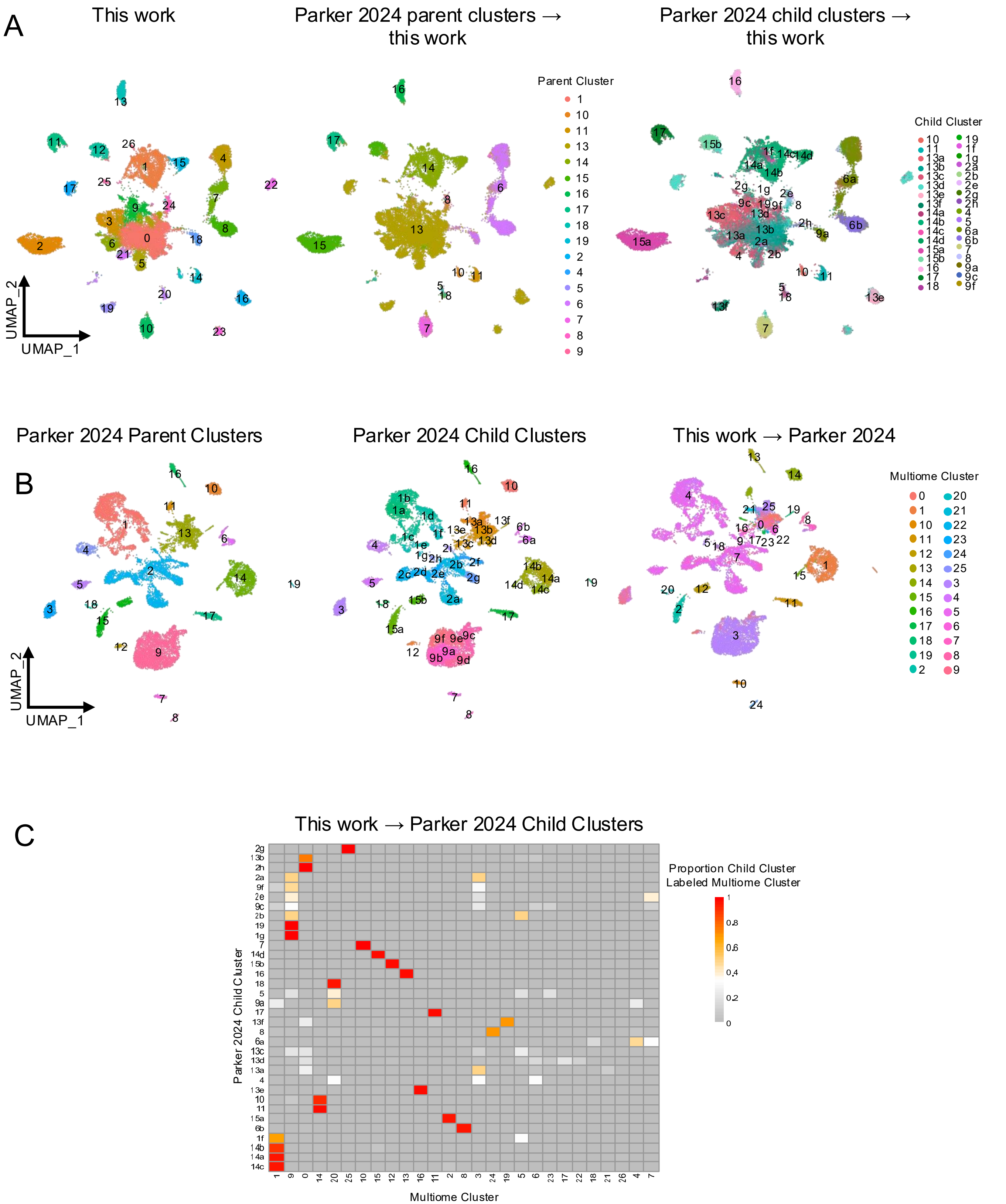
Anatomical Region H&E Dissection Plane



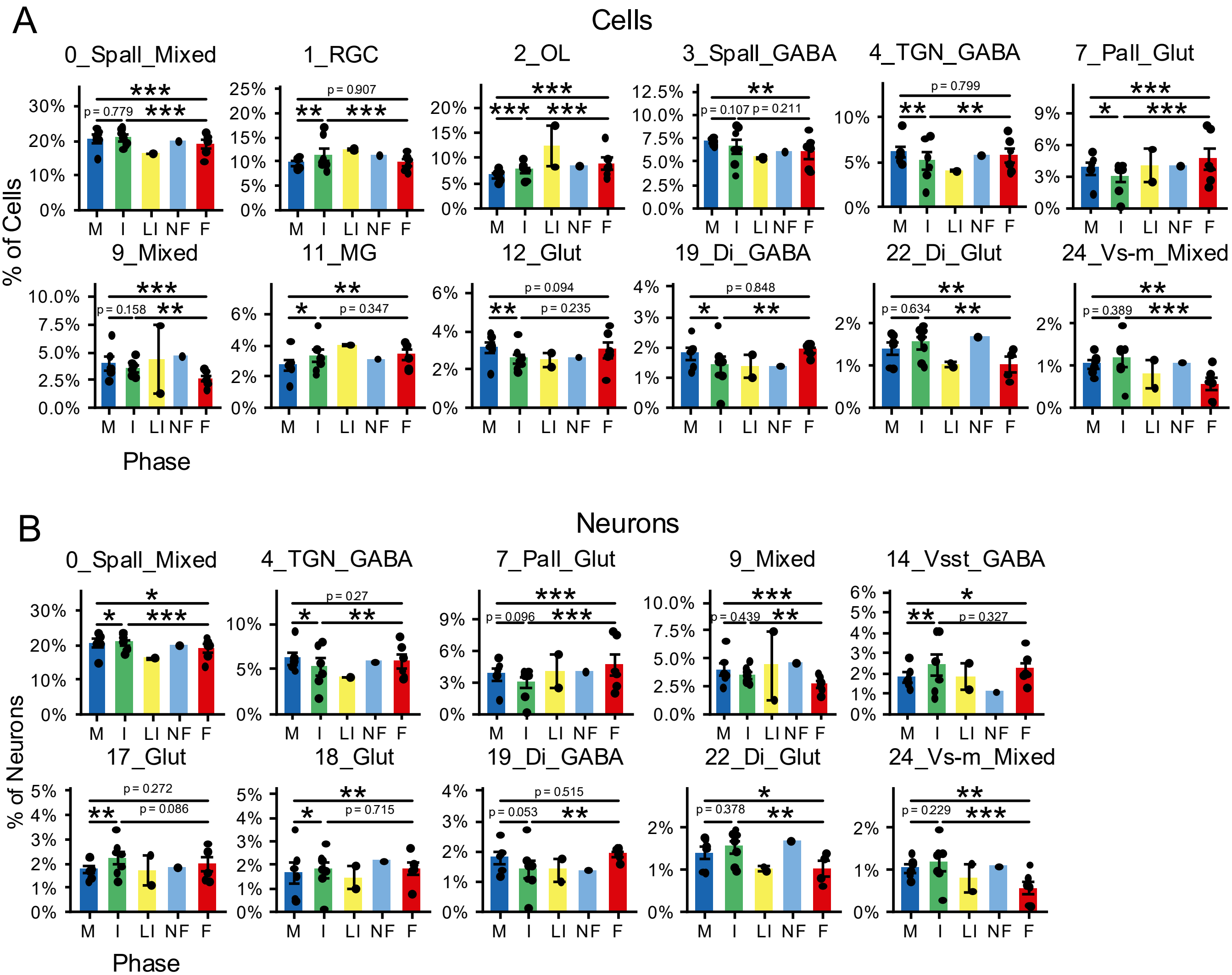
Supplementary Fig.3. Drawings of each section sequenced with VisiumHD with neuroanatomical regions labeled. The middle column displays the unlabeled hematoxylin and eosin stain and the right column shows a small brain from the side with a black line indicating the sectioning plane for the given section.



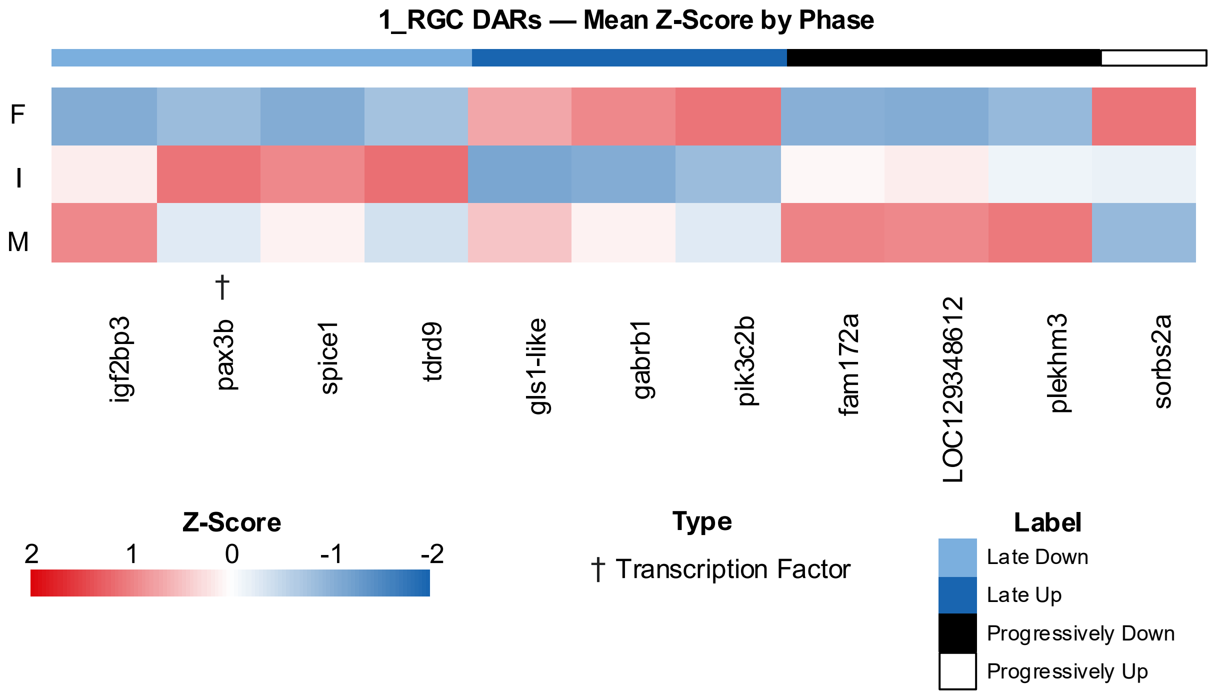
Supplementary Fig.4 Heatmap of Parker 2024 parent clusters mapped onto the Visium HD data in this work. The color in the heatmap indicates the scaled proportion of cells from each parent cluster composing each neuroanatomical region.


A
This work
Parker 2024 parent clusters → this work
Parker 2024 child clusters → this work
Parent Cluster
Child Cluster
UMAP_2
UMAP_1
B
Parker 2024 Parent Clusters
Parker 2024 Child Clusters
This work → Parker 2024
Multiome Cluster
UMAP_2
UMAP_1
C
This work → Parker 2024 Child Clusters
Proportion Child Cluster
Labeled Multiome Cluster
Parker 2024 Child Cluster
Multiome Cluster

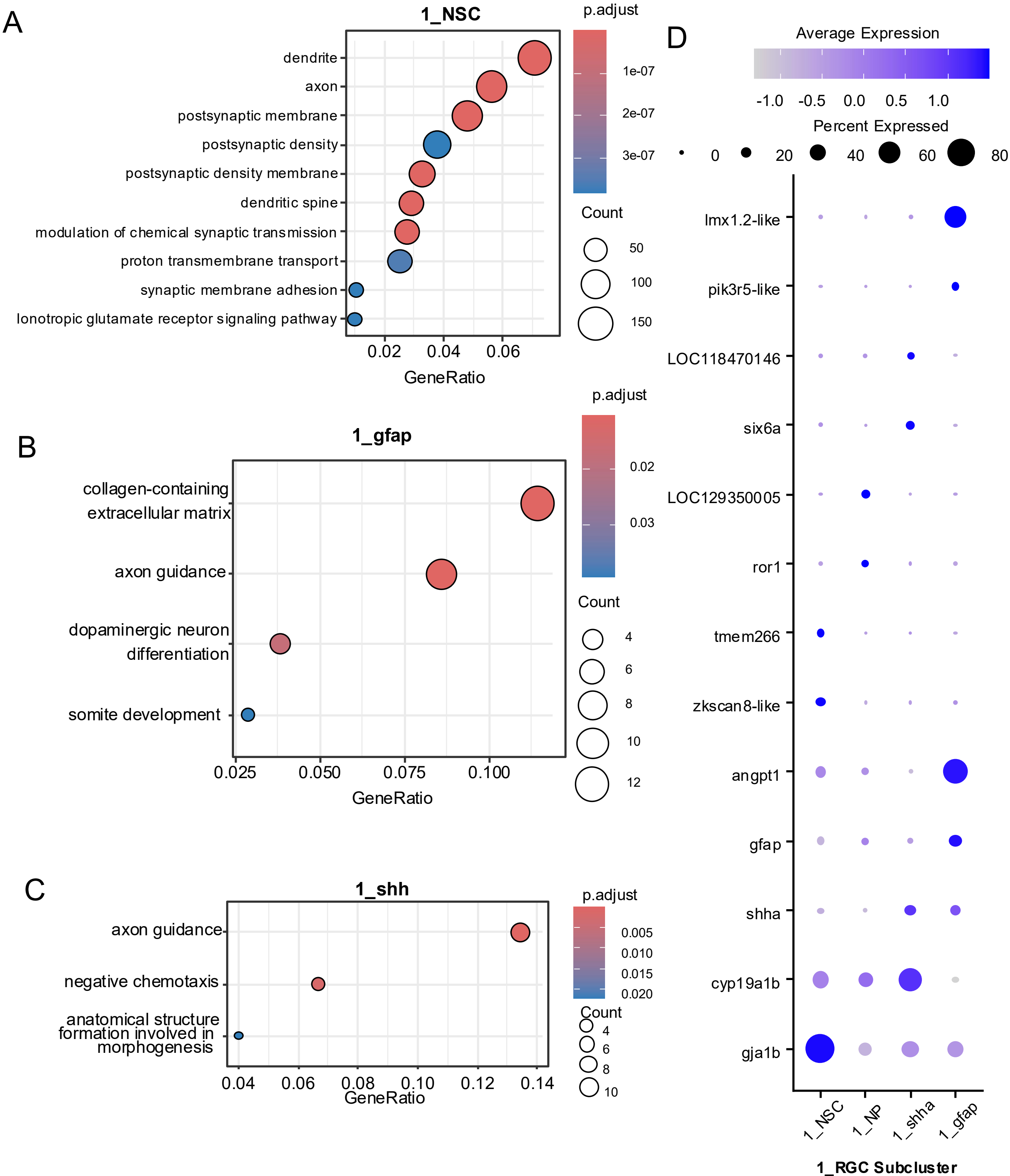
Supplementary Fig.5 Results of label transfer from Parker 2024 dataset. (A) Projection of Parker 2024 parent and child clusters onto the current work. (B) Clusters from the current work projected onto the Parker 2024 object. (C) Heatmap showing the proportion of each child cluster in the Parker 2024 labeled with the clusters from this work in a multiome to Parker 2024 child cluster mapping.



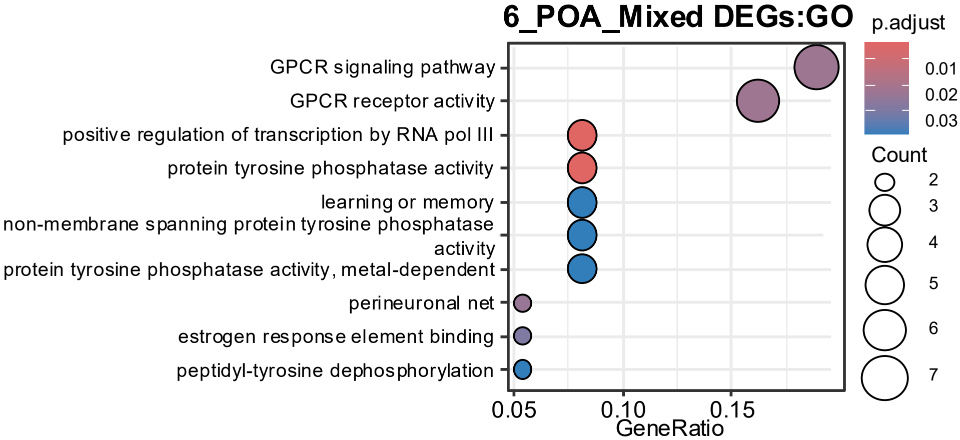
Supplementary Fig.6. (A) Percent of all cells belonging to each multiome cluster. (B) Precent of all neurons belonging to each multiome cluster. Only significant results are shown. Points indicate the mean for each individual. Asterisks: “*”: $p < 0.05$, “**”: $p < 0.01$, “***”: $p < 0.001$.



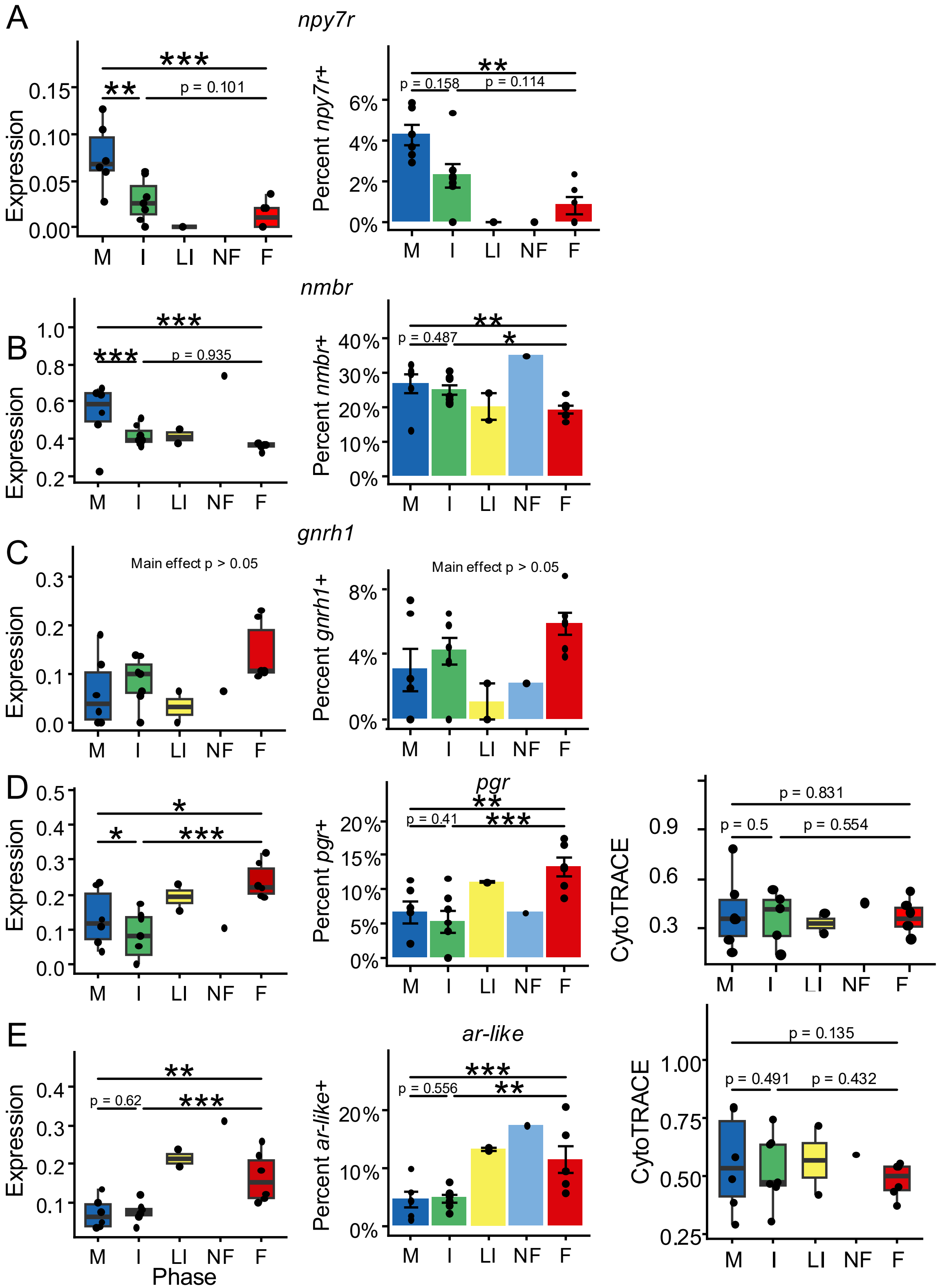
Supplementary Fig.7. DARs in 1_RGC. Cells are colored by z-score of the mean accessibility or expression for each phase. The top bar indicates the classification for how each gene or region changes sex, and the symbols below the matrix indicate what kind of gene is differentially accessible or differentially expressed. (B)


A
1_NSC
p.adjust
dendrite
axon
postsynaptic membrane
postsynaptic density
postsynaptic density membrane
dendritic spine
modulation of chemical synaptic transmission
proton transmembrane transport
synaptic membrane adhesion
Ionotropic glutamate receptor signaling pathway
0.02
0.04
0.06
GeneRatio
1e-07
2e-07
3e-07
Count
50
100
150
B
1_gfap
p.adjust
collagen-containing extracellular matrix
axon guidance
dopaminergic neuron differentiation
somite development
0.025
0.050
0.075
0.100
GeneRatio
0.02
0.03
Count
4
6
8
10
12
C
1_shh
p.adjust
axon guidance
negative chemotaxis
anatomical structure formation involved in morphogenesis
0.04
0.06
0.08
0.10
0.12
0.14
GeneRatio
0.005
0.010
0.015
0.020
Count
4
6
8
10
D
Average Expression
-1.0
-0.5
0.0
0.5
1.0
Percent Expressed
0
20
40
60
80
lmx1.2-like
pik3r5-like
LOC118470146
six6a
LOC129350005
ror1
tmem266
zkscan8-like
angpt1
gfap
shha
cyp19a1b
gja1b
1_NSC
1_NP
1_shha
1_gfap
1_RGC Subcluster

Supplementary Fig.8. Gene ontology (GO) analysis for the marker genes of (A) 1_NSC, (B) 1_gfap, and (C) 1_shha. In A-C, color indicates false discovery rate adjusted p value, point size indicates the number of genes annotated with that term, and “GeneRatio” indicates the proportion of genes annotated with that GO term divided by all annotated genes in our list. (D) Canonical and data-driven marker genes for each 1_RGC subcluster



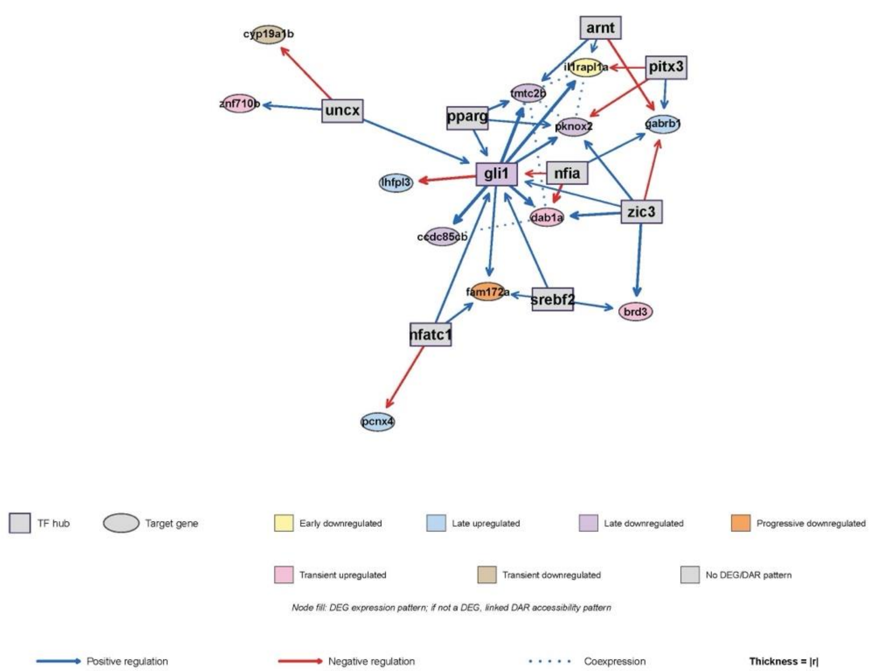
Supplementary Fig.9. GO analysis of all of the DEGs in 6_POA_Mixed. Ccolor indicates false discovery rate adjusted p value, point size indicates the number of genes annotated with that term, and “GeneRatio” indicates the proportion of genes annotated with that GO term divided by all annotated genes in our list.



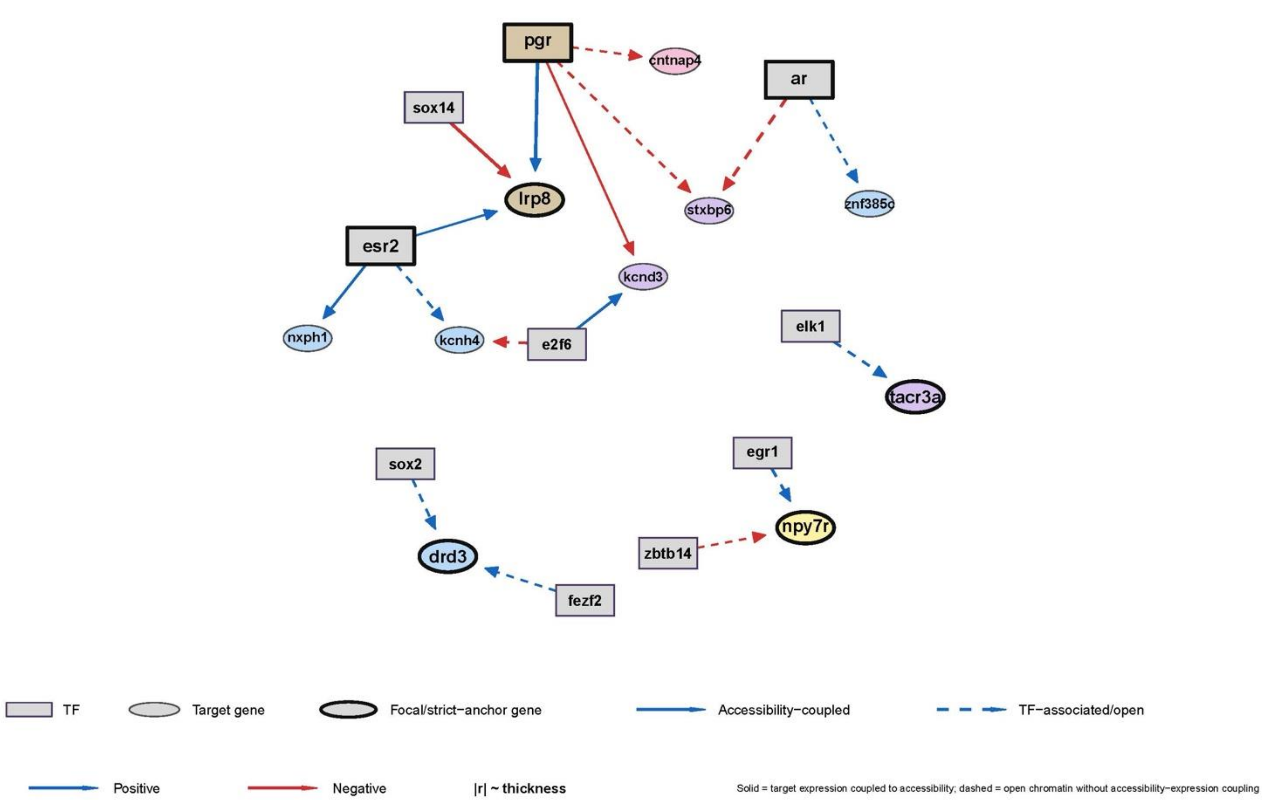
A
npy7r
Expression
Percent npy7r+
p = 0.101
p = 0.158
p = 0.114
B
nmbr
Expression
Percent nmbr+
p = 0.935
p = 0.487
C
gnrh1
Main effect p > 0.05
Expression
Percent gnrh1+
D
pgr
Expression
Percent pgr+
p = 0.41
CytoTRACE
p = 0.831
p = 0.5
p = 0.554
E
ar-like
Expression
Percent ar-like+
p = 0.62
p = 0.556
CytoTRACE
p = 0.135
p = 0.491
p = 0.432
M
I
LI
NF
F
Phase

Supplementary Fig.10. Expression (left) and percent of 6_POA_Mixed (right) expressing (A) *npy7r*, (B) *nmbr*, (C) *gnrh1.* Expression (left) and percent of 6_POA_Mixed (middle), and CytoTRACE score (right) of cells expressing (D) *pgr*, (E), *ar-like.* Points indicate the mean for each individual. Asterisks: "*": $p < 0.05$, "**": $p < 0.01$, "***": $p < 0.001$.

Cluster 1: Integrated Transcription Factor Regulatory Network



Cluster 6: Combined Reproductive + Accessibility−Coupled Regulatory Network



Supplementary Fig. 11 xx Justin please write